

Semi-Analytic Two-Dimensional Model for Supersonic Marshak Waves in Cylindrical Geometry

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TODO (Abstract draft)

INTRODUCTION

The experimental setup used for supersonic x-ray transport measurements is shown in Fig. 1 [7].

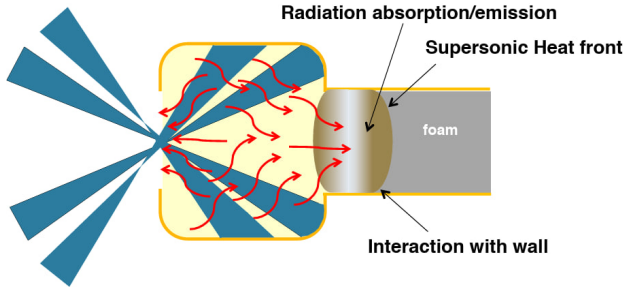


FIG. 1. Experimental setup adapted from Moore *et al.* [7].

GOVERNING EQUATIONS

MODELING THE HEAT WAVE

In this work, we develop a semi-analytic model for supersonic Marshak-wave propagation in cylindrical geometry that extends the Hammer–Rosen one-dimensional framework [1] to experimentally relevant finite-tube configurations. The model includes four principal physical mechanisms that control front propagation: (1) the correct incoming radiative flux at the foam boundary (Marshak boundary condition), (2) energy loss to wall heating, (3) wall ablation and the associated modification of the effective propagation channel, all presented in Refs. [8, 9], and (4) finite-diameter transport effects represented through an effective mean free path (Ref. [12]). The first mechanism is primarily longitudinal (1D), whereas the remaining mechanisms are wall-coupled and therefore intrinsically multidimensional. This separation motivates our staged construction: a corrected 1D longitudinal model followed by a full 2D description of front curvature and observables, consistent with the bent-wave formulations in Refs. [10, 11].

Hammer and Rosen [1] solved Eq. (??) in one dimension using perturbation theory. The heat front position

is given by

$$x_F^2(t) = \frac{2+\varepsilon}{1-\varepsilon} C H^{-\varepsilon}(t) \int_0^t H(t') dt', \quad (1)$$

where $H(t) = T_s^{4+\alpha}(t)$, $\varepsilon = \beta/(4+\alpha)$, and

$$C = \frac{16 g \sigma_{\text{SB}}}{3f(4+\alpha)} \rho^{2-\mu-\lambda}. \quad (2)$$

Here T_s is the surface temperature at $x = 0$, which HR set equal to the hohlraum drive temperature T_D .

CONCLUSION

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